

The Problem

Indian banks lost **Rs. 71,543 Cr** to fraud in FY 2024-25 (RBI Annual Report). The infrastructure designed to prevent this is not underfunded; it is broken by design.

Who Is Affected and How Severely

- **Direct:** Union Bank's AML compliance teams process hundreds of false alerts daily. Genuine fraud rings are buried due to investigator fatigue.
- **Financial:** IBM's 2024 Financial Services benchmark finds a **277-day median gap** between fraud initiation and detection. By then, recovery is effectively impossible.
- **Operational:** Manual Suspicious Transaction Report (STR) preparation takes **4 to 6 hours per case**. STR quality varies across officers, reducing the intelligence value of every FIU-IND filing.
- **Regulatory:** Under PMLA 2002, PSBs face penalties for late or incomplete STR filings.

Why Current Approaches Fail

Rule-based AML systems at Indian PSBs operate on fixed thresholds, generating 95%+ false positive rates. Three fundamental failures result:

Failure Mode	Consequence
95%+ false positives	Investigators spend their day dismissing noise. Genuine fraud rings are buried.
Single-transaction scope	Multi-hop laundering is invisible to per-transaction rules. Funds move through 4 to 8 intermediate accounts with no single hop triggering a threshold.
Reactive, not adaptive	Rules are static. New fraud typologies require a deployment cycle before becoming detectable.
No cross-account correlation	Siloed systems cannot connect related accounts executing coordinated round-trips.

The Patterns That Rules Miss

- **Layering chains:** Funds traverse 4 to 8 intermediate accounts, each forwarding more than 70% of received value within 48 hours. No single hop is suspicious; the chain is.
- **Round-trip (circular flow):** Money exits Account A, passes through 3 to 6 accounts, and returns within 72 hours. Net movement is zero. The purpose is to fabricate a paper trail of legitimate activity.
- **Structuring:** Multiple transactions just below Rs. 10 lakh are split across beneficiaries on the same day to evade CTR thresholds.

The Solution: TRACE.ai

TRACE.ai is a graph-native AML intelligence platform. Every banking transaction becomes a directed edge in a dynamic graph. Fraud patterns emerge as structural properties of the graph, detectable only at the network level.

Core Architecture: Four Parallel Detection Signals

Signal (weight)	How It Works
Pattern Matcher (30%)	Five AML typology detectors on a NetworkX MultiDiGraph: circular flow, layering, structuring, mule fan-in/out, dormant burst.
Graph Classifier (30%)	XGBoost on 11 graph-structural features (PageRank, betweenness, velocity, KYC risk). AUC above 0.99 on synthetic test data.
Online Anomaly (20%)	River HalfSpaceTrees per-account behavioral baselines with ADWIN drift detection, sub-millisecond latency, no batch retraining.
Compliance Engine (20%)	YAML-defined rules encoding RBI Master Directions, PMLA 2002, and FIU-IND STR triggers. Hot-reloadable without a system restart.

Fused thresholds: CRITICAL ≥ 0.85 , HIGH ≥ 0.70 , MEDIUM ≥ 0.50 .

Intelligence Layer

- **SHAP Explainability:** TreeExplainer attribution on every alert, showing the top-5 graph features that drove the risk score.
- **Gemini LLM Explainer:** Structured evidence sent to Gemini 2.5 Flash for a 2 to 3 sentence plain-English explanation per alert.
- **NL Investigation Copilot:** Officers ask in English (e.g., “Show circular flows above Rs. 5 lakh in the last 7 days”); the system dispatches to 6 pattern-specific graph query handlers.
- **Auto-STR Generator:** One click produces a FIU-IND compliant 8-section A4 PDF. Generation takes under 5 minutes (previously 4 to 6 hours per case).

Built vs. Planned

Capability	Demonstrable in POC	Planned (Not Yet Built)
Graph engine	NetworkX (542 nodes, 5,049 edges)	Neo4j GDS for millions of accounts
Pattern detection	5 AML typologies, live	15 to 20 additional typologies
ML classifier	XGBoost, AUC > 0.99 (synthetic)	Full TGN with real-bank data
Online anomaly scorer	River HST + ADWIN, under 1 ms	Kafka for real-time CBS stream
Compliance engine	YAML hot-reload, RBI/PMLA/FIU-IND	Auto-update from RBI circular API
Explainability	SHAP + Gemini per alert	Counterfactual explanations
STR generator	8-section FIU-IND PDF in under 5 s	Digital signature, submission channel
Dashboard	React 3D force-graph, live alerts	Multi-branch, multi-bank view
Data source	IBM AMLSim (100% synthetic)	Union Bank CBS API integration
Authentication	None (POC only)	LDAP/SSO for compliance teams

Measured Impact

Metric	Before TRACE.ai	With TRACE.ai
False positive rate	95%+ (rule-only)	Target below 15%
STR preparation time	4 to 6 hours	Under 5 minutes
Detection scope	Single-transaction rules	Multi-hop graph (depth 6)
ML adaptation	Quarterly batch retraining	Real-time, zero downtime

Live prototype: <https://trace-ai-4xnpj5ovp4a-uc.a.run.app>

Repository: <https://github.com/wildcraft958/namofans-trace-ai>